QUANTUM MACHINE LEARNING · PROJECT PROPOSAL

GNN vs Quantum-Inspired GNN vs Quantum GNN

A comparative analysis for molecular property prediction on the QM9 dataset

Rohit · October 2026

One prediction task, three model families

CLASSICAL

Graph Neural Network

Atoms are nodes, bonds are edges. Message passing (GCN / GIN / MPNN) learns a vector for the molecule, then a regressor predicts the property.

QUANTUM-INSPIRED

QI-GNN

Still runs on a classical computer, but adds physics: edge embeddings from interatomic distances and directions, as in quantum-chemistry interactions.

QUANTUM

Quantum GNN

A hybrid model: features are encoded into qubits and a variational quantum circuit does part of the message passing or readout (simulated).

Fair comparison: same molecules, same split, same target, same metric (MAE in eV), plus parameter count and training time.

02 · DATASET

QM9: small organic molecules with DFT-computed properties

133,885
molecules in the raw file

130,745
after cleaning

≤ 9
heavy atoms (C, N, O, F) plus H

12
quantum-chemical targets

Group	Columns	Behaviour
Electronic	homo, lumo, gap, mu	Intensive (size-independent)
Size / shape	alpha, r2, A, B, C	Mostly extensive
Thermochemical	u0, u298, h298, g298, zpve, cv	Extensive (grow with size)

03 · TARGET

Predict the HOMO–LUMO gap, in eV

Why this target

- **Intensive:** it does not grow with molecule size, so a model cannot score well just by counting atoms.
- **Well-behaved:** almost symmetric (skew 0.00), mean 6.86 eV, std 1.28 eV, only 0.07% IQR outliers.
- **Meaningful:** controls reactivity, colour and electronic behaviour (OLEDs, solar cells).
- **Comparable:** two of my three papers report gap MAE (≈ 0.12 eV), so results can be benchmarked.

Never used as inputs

homo and lumo: gap = lumo − homo exactly, so using them is target leakage.

Every other DFT property (mu, alpha, energies, A, B, C) is also excluded. Inputs come only from the molecule's structure.

Optional second target: dipole moment μ, to compare with paper 3.

Inputs: the molecule as a graph

Feature group	What it contains	Used by
Node (atom)	Element one-hot (H, C, N, O, F), degree, hybridisation (sp, sp^2, sp^3), aromatic, in ring, number of H, formal charge	All three models
Edge (bond)	Bond type one-hot (single, double, triple, aromatic), conjugated, in ring	All three models
3D geometry	Interatomic distance (radial-basis expanded) and unit direction vector, from the DFT coordinates	QI-GNN
Qubit encoding	Heavy-atom graph: at most 9 nodes, so at most 9 qubits. Node features compressed and angle-encoded as RY(θ), θ in [0, π]	Quantum GNN
Baseline	165 cleaned RDKit descriptors and a 1,024-bit Morgan fingerprint; 8 PCA components as 8 qubit angles (already saved)	Sanity check

The data is clean, but the chemistry is imbalanced

Data quality

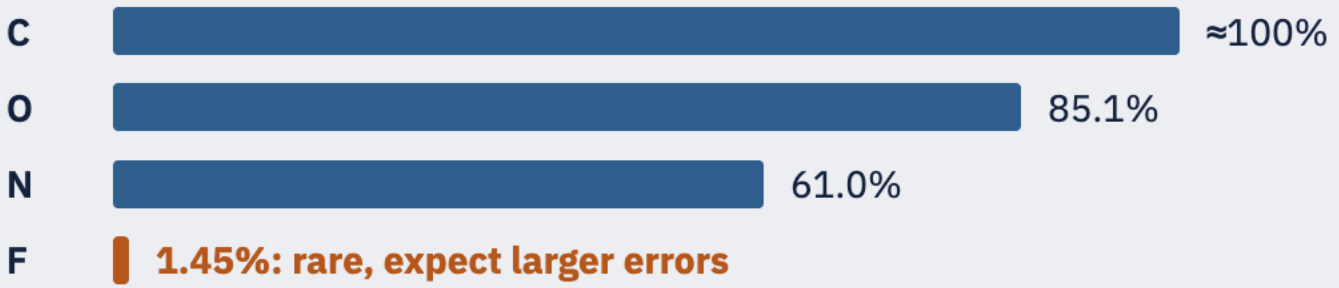
0 missing values in any target

83 duplicate SMILES in the raw file, removed by canonical SMILES

3,054 uncharacterised molecules removed (failed QM9's geometry check)

Invalid SMILES checked with RDKit

Share of molecules containing each element



83%
have exactly 9 heavy atoms

67%
have 1 or 2 rings (10% none)

17%
have an aromatic ring

Using GNN property predictors as molecule generators

Therrien, Sargent and Voznyy · Nature Communications 16:4301 (2025)

Classical GNN

What they did

Trained a simple graph convolutional network (GCN) on QM9 to predict the HOMO–LUMO gap from 2D graphs only. Then they froze the weights and optimised the input graph to generate new molecules with a chosen gap, checked with DFT.

0.12 eV

GCN test MAE on gap (PaiNN with 3D: 0.048)

0.96 eV

GCN MAE on 1,617 new molecules (PaiNN: 1.16)

For my project

Gives the classical baseline to beat (≈ 0.12 eV with a 2D GCN), and shows that QM9 test scores overstate accuracy on unseen molecules, so I also need a scaffold split.

Molecular property prediction using quantum-inspired GNNs

Chanana · Springer Proceedings in Physics 369, IC-AMSI 2026

Quantum-inspired GNN

What they did

Proposed QIGNN: message passing with learned edge embeddings built from interatomic distances and direction vectors, and a multi-head regressor that predicts many QM9 properties at once.

Property	MAE
HOMO–LUMO gap	0.1186 eV
HOMO	0.0803 eV
LUMO	0.0960 eV
Dipole moment μ	0.1609 D

For my project

The reference design and target number for my quantum-inspired model. Note that "quantum-inspired" here means physics-informed geometry, not qubits.

Dipole moment prediction with GNNs and physics-informed models

Li, Bian, Yang, Hu, Xu, Zhou and Hu · Chemical Papers 80 (2026)

GNN vs physics-informed

What they did

Compared six models for dipole moment: three GNNs (MPNN, GAT, GIN) and three PhysNet variants. Trained on QM9, then tested on 50 Traditional Chinese Medicine molecules (23 not in QM9).

PhysNet
most accurate on QM9

GAT 1.48×
smallest error increase on new molecules

For my project

A template for a fair multi-model comparison on QM9. It suggests trying GAT and GIN as classical baselines and reporting out-of-distribution error, not only test MAE.

11 · doi.org/10.1007/s11696-026-04657-0 · results taken from the abstract (full text is paywalled)